TUTORIAL

Radial distribution functions

> Draft, 2026-10-21. The figures were made with the current development version of EasyHybrid (they show the new g(r) plot of the RDF window, not the one in 3.1.3).

This tutorial computes two radial distribution functions from a short molecular dynamics run of chignolin (PDB 1UAO, a 10-residue β-hairpin) in a box of water:

1. water oxygen – water oxygen, the classic g(r) of liquid water;
2. carboxylate oxygens of Asp3 and Glu5 – water oxygen, the hydration shell of the charged side chains.

The whole tutorial takes about 5 minutes; the MD run itself about 2.5 minutes on one CPU core.

What g(r) is

g(r) is the density of B atoms found at distance r from an A atom, divided by the average density of B in the box. g(r) = 1 means "as in the bulk"; a peak means a preferred distance (a solvation shell); g(r) = 0 means no pairs at that distance.

The window also gives the running coordination number N(r), the average number of B atoms within r of an A atom. N(r) at the first minimum of g(r) is the number of neighbours in the first shell.

g(r) needs a periodic box: the density is taken from the box volume of each frame, and distances use the minimum image.

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1. Load the solvated system

File → Open and choose `examples/tutorials/04_md_basics/1UAO_OPLS_folded_solvated.pkl` from the EasyHybrid folder.

Atoms	2135 (chignolin, 665 water molecules, 2 Na^+)
Force field	OPLS-AA
Box	cubic, 28 Å, periodic boundary conditions
Nonbonded model	cut-off

The status bar at the bottom of the main window shows these properties (`force field: OPLS` , `PBC: True` , `symmetry: Cubic`).

2. Run a short MD simulation

Simulate → Molecular Dynamics. Fill in the window as in Figure 1:

Field	Value	Why
Initial Coordinates	`0_coords`	the structure just loaded
Integrator	Langevin Dynamics	keeps the temperature at 300 K
Number of Steps	10000	10 ps with a 1 fs step
Temperature (Start)	300 K	
Time Step (ps)	0.001	1 fs (pDynamo has no SHAKE)
Log Frequency	500	one line in the log every 0.5 ps
Collision Frequency	25.0 ps^{-1}	Langevin friction

Field	Value	Why
Save Trajectory	on, name `1UAO_md_300K`	
Format	pDynamo / pkl	
Trajectory Frequency	100	one frame every 0.1 ps, 101 frames

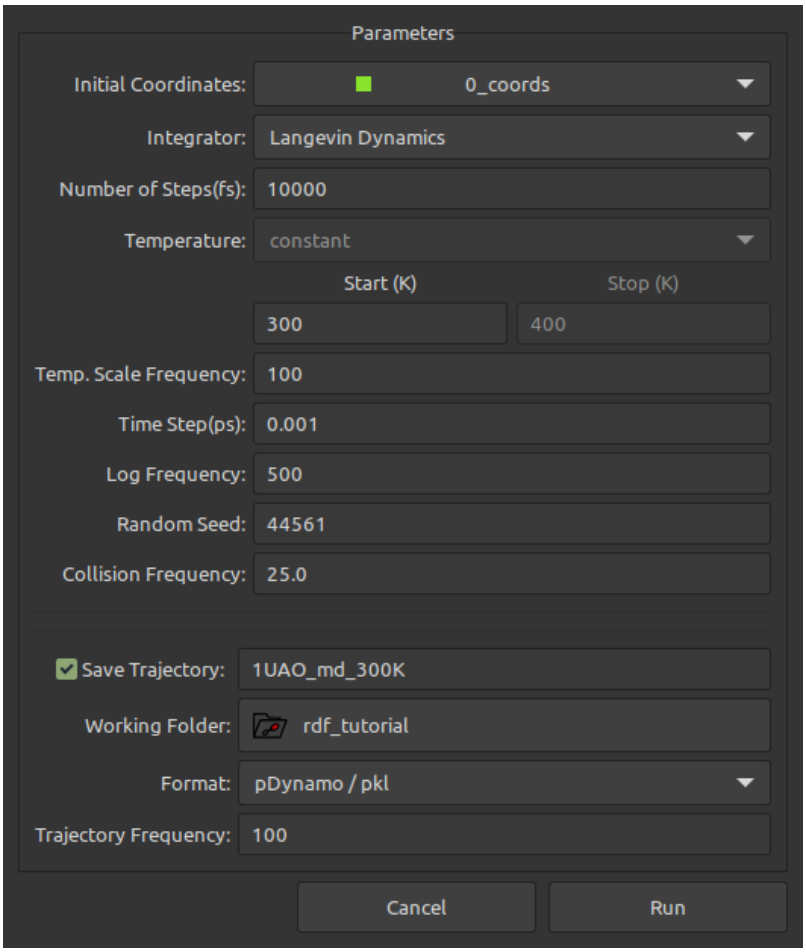


Figure 1. Molecular Dynamics window for the 10 ps run.

Click **Run**. The job runs in the background; its progress is in the Process Manager. In our run the average temperature was 296 ± 5 K.

> **10 ps is enough for this tutorial, not for a study.** The run starts from the stored structure without equilibration. For publishable g(r), equilibrate first and analyse a longer production run (hundreds of ps or more), with more frames.

3. Import the trajectory

When the job finishes, the **Finished!** dialog appears (Figure 2). Click **Import...**: the **Import Data** window opens already filled in with the job's trajectory (`1UAO_md_300K.ptGeo` , format *pkl folder – pDynamo trajectory*) and its log (Figure 3). Click **import**.

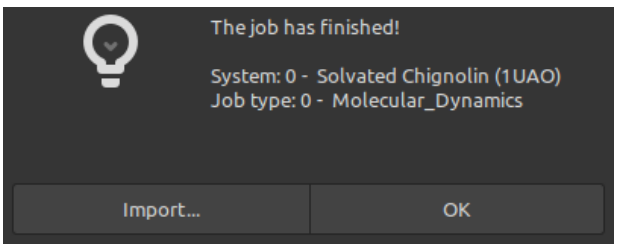


Figure 2. The Finished! dialog. Import... opens the Import Data window for this job.

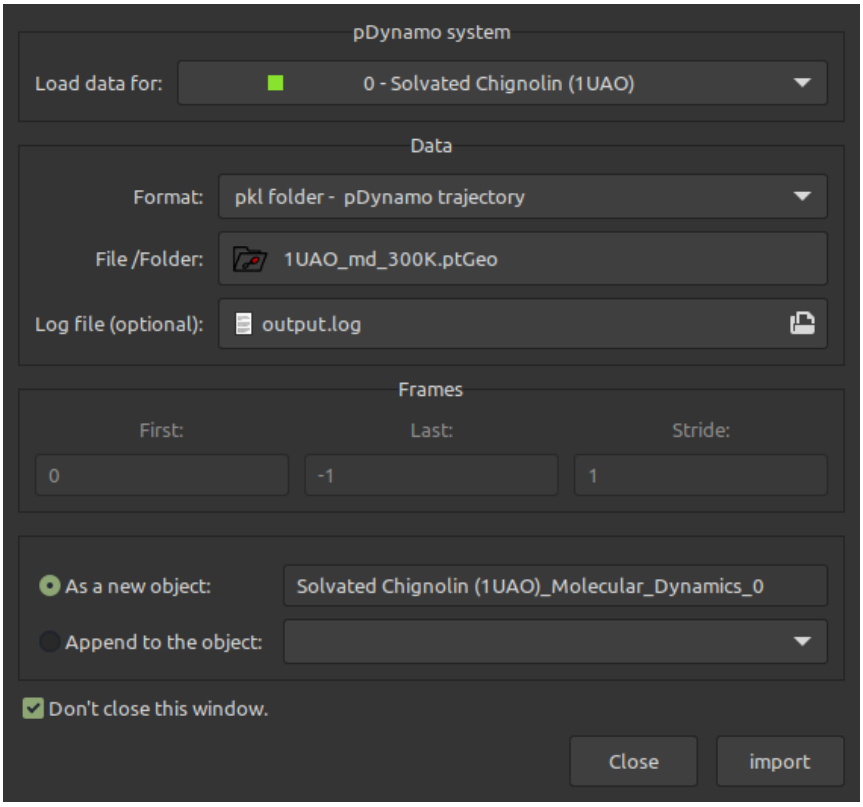


Figure 3. Import Data, pre-filled with the MD trajectory.

A new object, `…_Molecular_Dynamics_0`, appears under the system in the object list, with 101 frames (column F). Its frames keep the periodic box, which the RDF needs. (The MD run also adds an object with only its last frame, `0_Dynamics_0`; it is not used here.)

4. Define the two groups of atoms

The RDF window takes its groups A and B from the system's selection list (**Selection → Selections and Restraints**). The quickest way to select every water oxygen is the EasyHybrid Terminal.

1. **View → EasyHybrid Terminal**.
2. Type `list` to see the object numbers. The trajectory is object 2 here.
3. Select the water oxygens: `select obj=2 resn=HOH name=O` The terminal answers `665 atoms` (Figure 4).
4. **Selection → Send to Selection List**. The selection is stored as `sel_0`.
5. Select the four carboxylate oxygens (Asp3 OD1/OD2 and Glu5 OE1/OE2) and send them too (`sel_1`): `select obj=2 resn=ASP,GLU name=OD1,OD2,OE1,OE2`
6. Give the selections meaningful names: in **Selection → Selections and Restraints**, right-click a selection, choose **Rename**, type the new name and press Enter. We used `water_O` and `carboxylate_O` (Figure 5).
7. Type `deselect` in the terminal to clear the highlight in the 3D view.

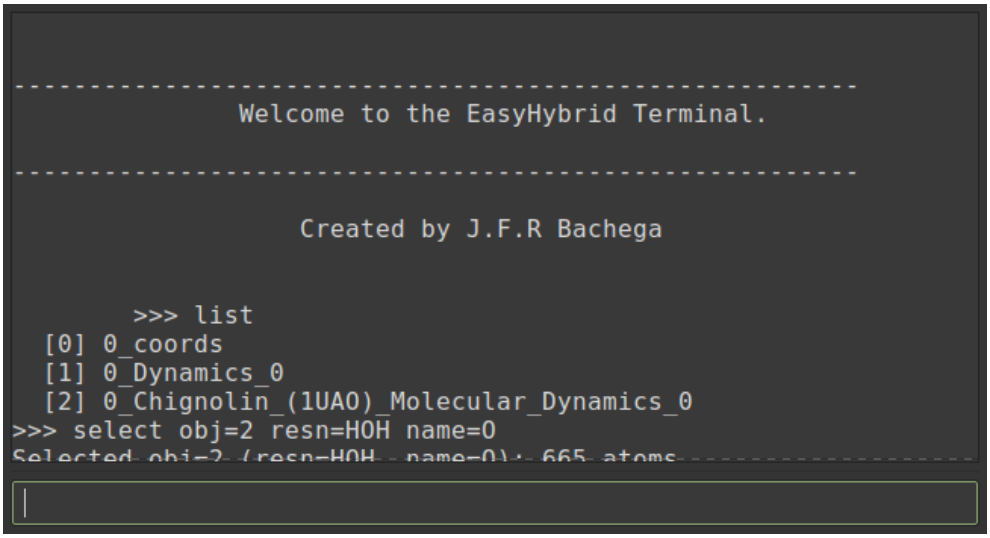


Figure 4. Selecting the 665 water oxygens in the EasyHybrid Terminal.

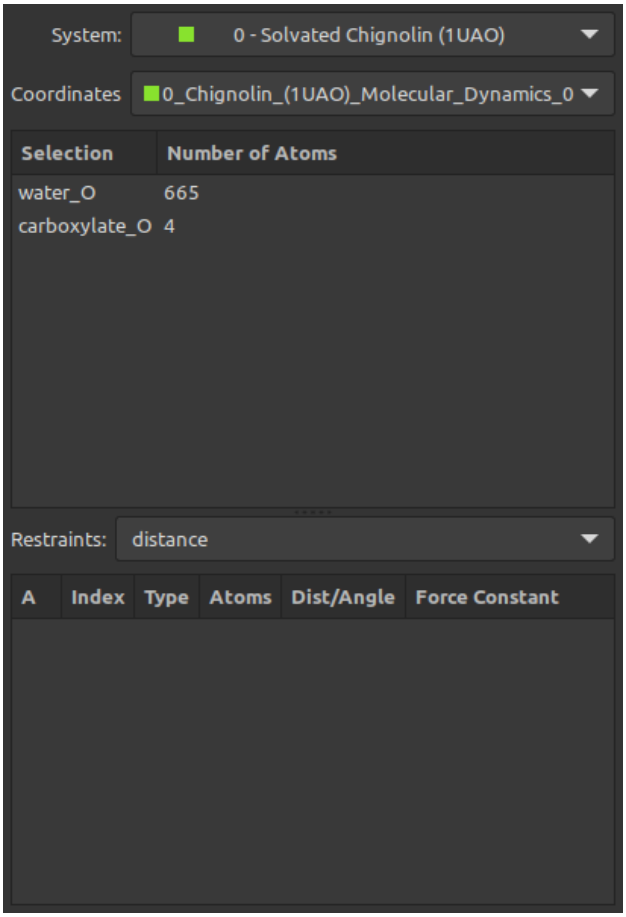


Figure 5. The two groups in the selection list, after renaming.

The terminal filters combine (AND): `chain=`, `resi=` (ranges such as `10-20`), `resn=`, `name=` and `symbol=` (element), each accepting comma-separated values, plus `not_…` versions to exclude atoms. `help select` lists them.

5. Water oxygen – water oxygen g(r)

Analysis → RDF Analysis (g(r)) (Figure 6):

Field	Value
System / Object	the solvated chignolin / the MD trajectory (`…_Molecular_Dynamics_0`)
Group A	`water_O`
Group B	`water_O` (the same group: each atom is not counted as its own neighbour)

Field	Value
r max	8 Å (keep it below half the box length, 14 Å here)
Bin width	0.05 Å
Frame range	all frames (0 to −1), step 1
Plot using EasyPlot	on

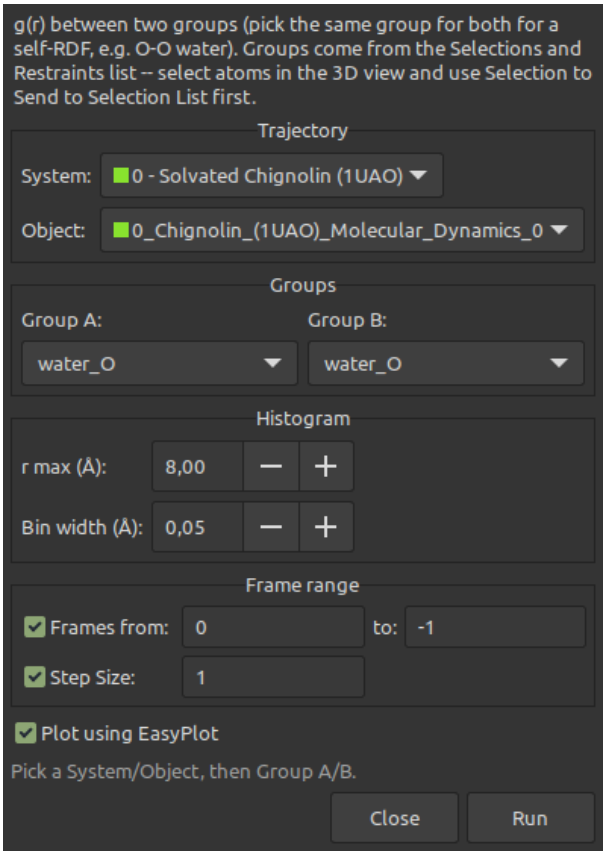


Figure 6. RDF Analysis window for the water O–O g(r).

Click **Run**. Two windows open: a table with r, g(r) and N(r) for every bin (it can be saved as text), and the plot (Figure 7). Moving the mouse over the plot shows the values under the cursor; **Export PNG…** saves the figure (Figure 8 was made this way).

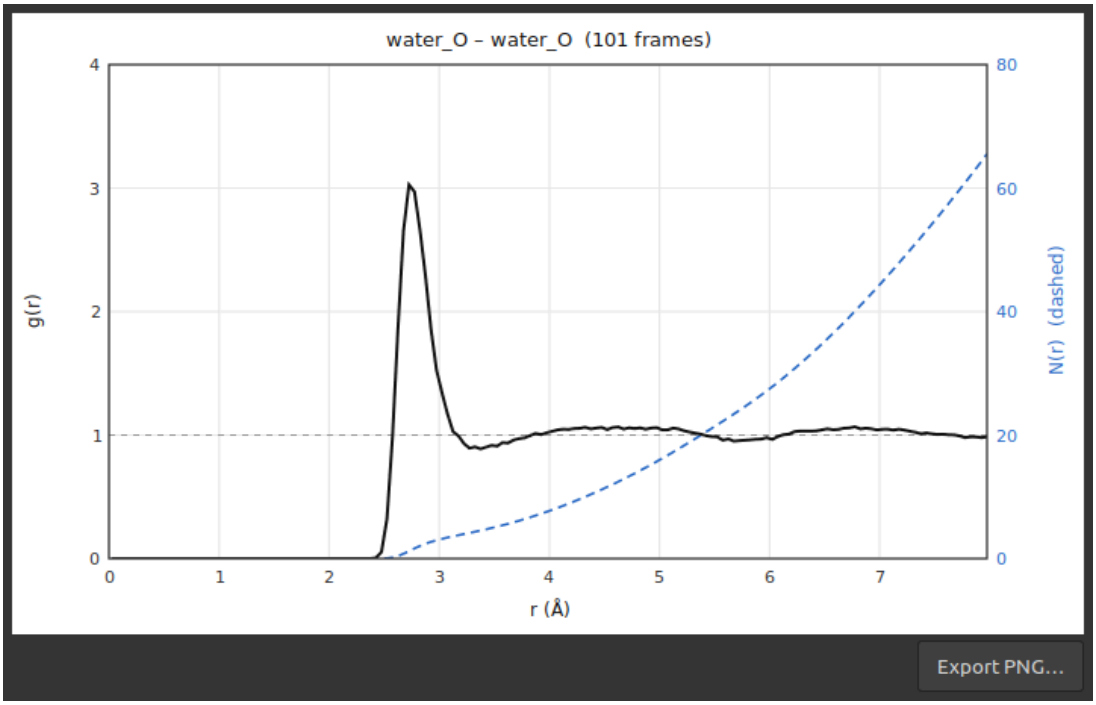


Figure 7. The g(r) plot window: g(r) (solid, left axis) and N(r) (dashed, right axis).

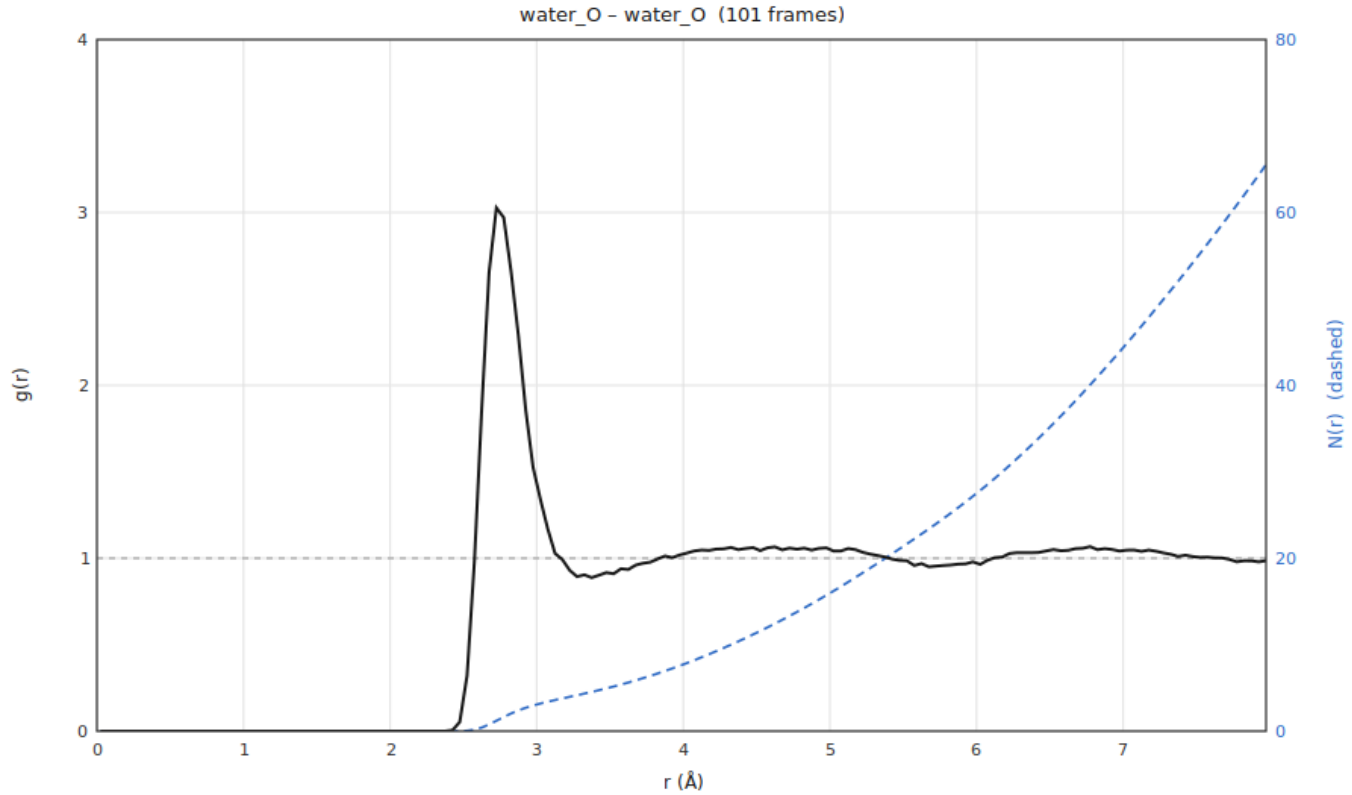


Figure 8. Water oxygen – water oxygen g(r) over 101 frames (10 ps).

What the curve shows:

	This run	Liquid water, experiment
First peak	2.73 Å, g = 3.0	2.8 Å, g ≈ 2.5
First minimum	3.38 Å	≈ 3.4 Å
N(r) at the first minimum	4.5	≈ 4.3–4.7

Each water molecule has about four nearest neighbours, the hydrogen-bonded tetrahedral shell. The first peak is higher than in experiment, as is usual for TIP3P-type water models; beyond 4 Å, g(r) oscillates around 1.

6. Carboxylate oxygen – water oxygen g(r)

In the same window change **Group A** to `carboxylate_O` (keep Group B = `water_O`) and click **Run** again (Figure 9).

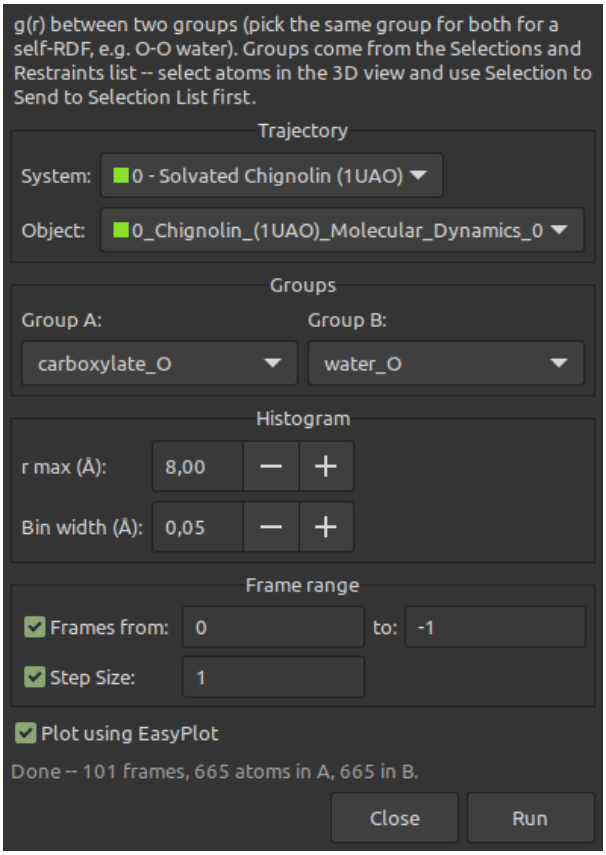


Figure 9. Group A changed to the carboxylate oxygens.

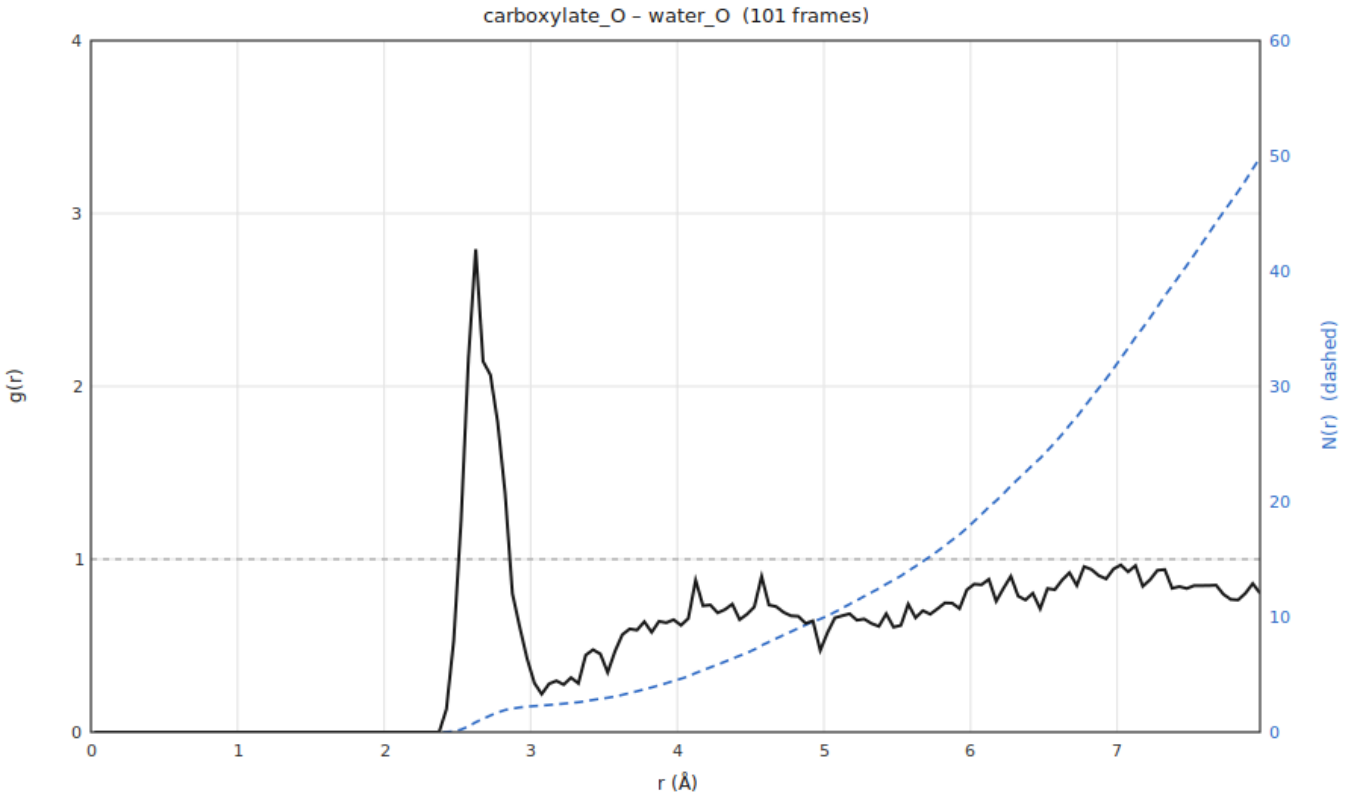


Figure 10. Carboxylate oxygen (Asp3, Glu5) – water oxygen g(r).

	This run
First peak	2.63 Å, g = 2.8
First minimum	3.08 Å (g = 0.2)
N(r) at the first minimum	2.3 water molecules per carboxylate oxygen

The first peak is closer than in bulk water (2.63 against 2.73 Å) and is followed by a deep minimum: the waters hydrogen-bond strongly to the charged carboxylate and form a well-defined shell. Beyond the first shell g(r) stays below 1, because part of the space around a side-chain oxygen is taken by the peptide itself, not by water.

The curve is noisier than the water O–O one: group A has 4 atoms instead of 665. More frames (a longer run) smooth it.

Notes

- **Same group in A and B** (self-RDF): an atom is not counted as its own neighbour. Two different groups that only partly overlap are not treated specially.
- **r max** must stay below half the shortest box length, or the minimum-image distances are wrong.
- **Frames from / Step Size**: skip the first frames (equilibration) or use every n-th frame for a long trajectory.
- **No box**: if the object has no periodic cell, the window says so; g(r) cannot be normalised without a volume.
- **Other force fields**: the analysis only reads coordinates and the box, so it works the same for AMBER, CHARMM, OPLS or DYFF trajectories, as long as the imported frames carry the periodic box.

EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3